Md. Najmul Hossain Nur

CSE Undergraduate, United International University, Dhaka, Bangladesh

mnajmulhossainnur@gmail.com — +880 1743094595 — Dhaka, Bangladesh — LinkedIn — GitHub — Portfolio

Research Interests

Machine Learning • Computer Vision • Autonomous Systems & UAVs • Embedded Systems & Robotics • Full-Stack Web Development • UI/UX Design

Education

United International University, Dhaka, Bangladesh *2022 — Expected July 2027*

Bachelor of Science in Computer Science and Engineering — Current CGPA: 2.87/4.00

Collectorate School & College, Rangpur *2019 — 2021*

Higher Secondary Certificate (HSC), Science

Lions School and College, Rangpur *2017 — 2019*

Secondary School Certificate (SSC), Science

Research Experience

AgriML-BD — Crop Yield Prediction and Recommendation Using Machine Learning for Bangladesh Agriculture

- Ongoing undergraduate research project supervised by Ms. Sadia Islam, Assistant Professor, Department of CSE, UIU, focused on machine learning-based crop recommendation and yield prediction for Bangladeshi agriculture.

Robotics & UAV Experience

Uro Bangladesh — Robotics Startup & Research Organization *Team Member*

- Contributed as a team member of **Uro Bangladesh** at the **TEKNOFEST 2026 International UAV Competition** (Rotary Wing), where the team secured **10th place globally and 2nd in Asia** among 113 finalist teams.

Projects

Autonomous Underwater Monitoring Submarine (AUMS) (GitHub) *Micro Lab · Team Leader*

Underwater monitoring robot with navigation control and sensor integration that automatically searches for a target and sends captured images to a website.

Tech Stack: Raspberry Pi 5, ESP32, sensor and motor integration.

SearchAR — Community Safety Platform (GitHub) *SE Lab · Team Leader*

AI-powered platform for missing-person detection, fire hazard detection, and crowdsourced crime reporting using deep learning and computer vision. The system includes role-based dashboards for public users, volunteers, police, and administrators.

Tech Stack: Deep Learning, Computer Vision, Figma (UI/UX).

Smart Waste Management System (GitHub) *Electronics Lab · Team Leader*

Bluetooth-controlled robotic waste collection system with ultrasonic detection, robotic arm and gripper control, and real-time LCD monitoring through an MIT App Inventor application.

Tech Stack: Arduino Mega 2560, HC-05, HC-SR04, servo motors, 16x2 LCD.

Sohaj Jogan — Micro E-commerce Platform (GitHub) *Web Project · Team Leader*

Location-based shopping platform connecting buyers with nearby local shops through live inventory, delivery tracking, and in-app messaging.

Tech Stack: HTML, CSS, JavaScript, PHP, MySQL, Leaflet.js.

UNO Night — JavaFX Multiplayer Game (GitHub) *AOOP Lab · Team Leader*

Real-time multiplayer UNO game featuring four game modes, in-game currency and shop, a clan system, and Wi-Fi/LAN-based socket networking.

Tech Stack: Java, JavaFX, socket programming.

Restaurant Management System (GitHub) *Web Lab · Team Project*

PHP/MySQL-based platform for managing restaurant operations, including table booking, kitchen order tracking, and role-based dashboards for administrators, chefs, waiters, and customers.

Tech Stack: PHP, MySQL.

Design / UI-UX

- Designed and prototyped SearchAR mobile and web application flows in Figma, including emergency rescue interfaces, SOS navigation, and dashboard layouts — Figma Link.
- Designed the Restaurant Management System web prototype in Figma, focusing on clean dashboards and efficient order management — Figma Link.

Technical Skills

Programming: Java, C, C++, Python

Web Development: HTML, CSS, JavaScript, PHP, React, Node.js

Database: MySQL, SQL, PostgreSQL

Frameworks & Tools: JavaFX, Figma, Scene Builder, Git & GitHub

IoT / Embedded / Design: Arduino Mega, ESP32, Raspberry Pi 5, sensor and motor integration, Fusion 360

Languages: Bengali (First Language), English (Advanced, C1)

Achievements

- **10th Globally and 2nd in Asia** — TEKNOFEST 2026 International UAV Competition, Rotary Wing category, Malatya, Türkiye.
- **Champion** — UIU CSE Project Show, Spring 2025, Database Management Systems Project.
- **4th Place** — UIU CSE Project Show, Fall 2024, Advanced Object-Oriented Programming Project.
- **4th Place** — UIU CSE Project Show, Summer 2025, Systems Analysis and Design Project.
- **Top 10** — UIU CSE Project Show, Spring 2026, Software Engineering Project for SearchAR.